

Question 1 [Marks: 5]

Number of attacking Queen pairs, $c = 6$

	Q		Q	
Q				
	Q			
			Q	

Initial State

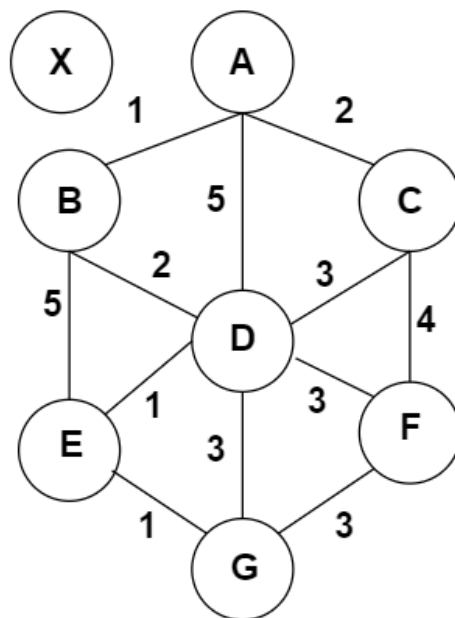
Number of attacking Queen pairs, $c = 0$

	Q			
			Q	
Q				
		Q		
				Q

Goal State

- A.** Observe the given states of a 5-Queen Problem. Now, **sketch a portion of the state space tree (branching factor = 2)** and showcase a scenario where hill climbing would get stuck in a **Local Maxima**. At each depth, you can select **only one Queen** and move it **Up, Down, Left, or Right**. [Note: You don't have to show the complete state space tree] **(5)**

Question 2 [Marks: 10]



States,n	h1(n)	h2(n)
A	5	5
B	2	3
C	4	5
D	2	2
E	1	1
F	2	3
X	??	??
G	0	0

- Determine/Estimate a heuristic value for node "X". Write a few words to explain the value determined for node "X" (2)
- Among the heuristic h1 and h2, perform A* search using heuristic that is better than the other. (4)
- Is it possible to determine the optimal path or sub-optimal path from node C to node E using any of the above heuristics values? If yes/no then please explain in a few words why. (2)
- If the cost from C to F and C to D becomes 3 and 2 respectively, then would the heuristic value of h2(B) be consistent? Show calculations (2)